T3 E_-4: general EFT1 RGEs

Matchete/RGBeta symbolic output in mathematical notation

E_-4

Additional quartics: 0, additional bilinears: 0, extra Yukawa: none.
General matching and Wilson coefficients are not computed in this symbolic report.

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{71{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{11{g_2}^3}{6}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = & \, 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = & \, \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} &= 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ &+ 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \frac{3}{2}\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij} \\ &- \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ &- \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

m_1^2

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} &= -12{g_2}^2 m_1^2 \\ &- 24{g_1}^2 m_1^2 \\ &+ m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\ &+ m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})\end{aligned}$$

m_2^2

$$\begin{aligned}16\pi^2 \frac{dm_2^2}{d\ln\mu} &= -12{g_2}^2 m_2^2 \\ &- 6{g_1}^2 m_2^2 \\ &+ m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\ &+ m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})\end{aligned}$$

λ_1

$$\begin{aligned}16\pi^2 \frac{d\lambda_1}{d\ln\mu} &= \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\ &- 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big) \\ &- 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\ &- 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big) \\ &+ 4\lambda_5\,(\lambda_5)^* \\ &+ \frac{9{g_2}^4}{4} \\ &+ {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right) \\ &- 9{g_2}^2\lambda_1 \\ &+ \frac{3{g_1}^4}{4} \\ &+ 12{\lambda_1}^2 \\ &+ 6{\lambda_{HS_1}^{(1)}}^2\end{aligned}$$

$$
\begin{aligned}
&+ 3{\lambda_{HS_1}^{(A)}}^2 \\
&+ 6{\lambda_{HS_2}^{(1)}}^2 \\
&+ 3{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = {} & 22{g_2}^4 \\
& + {g_1}^2\left(32{g_2}^2 - 48\lambda_{S_1}^{(1)}\right) \\
& - 24{g_2}^2\lambda_{S_1}^{(1)} \\
& + 192{g_1}^4 \\
& + 6{\lambda_{12}^{(1)}}^2 \\
& + 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
& + \frac{20{\lambda_{12}^{(\times)}}^2}{3} \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + 14{\lambda_{S_1}^{(1)}}^2 \\
& + 8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
& + 16{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = {} & 22{g_2}^4 \\
& + {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_2}^{(1)}\right) \\
& - 24{g_2}^2\lambda_{S_2}^{(1)} \\
& + 12{g_1}^4 \\
& + 6{\lambda_{12}^{(1)}}^2 \\
& + 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
& + \frac{20{\lambda_{12}^{(\times)}}^2}{3} \\
& + 4{\lambda_{HS_2}^{(1)}}^2 \\
& + 14{\lambda_{S_2}^{(1)}}^2 \\
& + 8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
& + 16{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = \lambda_{HS_1}^{(1)}(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_1}^{(A)})
$$

$$
\begin{aligned}
&+ 4\lambda_5 \left(\lambda_5\right)^* \\
&+ 6{g_2}^4 \\
&- \frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 12{g_1}^4 \\
&- \frac{51{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ \lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 6{\lambda_{HS_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_2}^{(1)}(6\,\mathrm{Tr}\Big(Y_d\ \left(Y_d\right)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ \left(Y_e\right)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ \left(Y_u\right)^\dagger\Big) + 4\lambda_{S_2}^{(A)}) \\
&+ 4\lambda_5 \left(\lambda_5\right)^* \\
&+ 6{g_2}^4 \\
&- \frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 3{g_1}^4 \\
&- \frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2} \\
&+ \lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ 6{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = {} & 2\lambda_5 \left(\lambda_5\right)^* \\
&+ \frac{63{g_2}^4}{5} \\
&- 24{g_2}^2\lambda_{12}^{(1)} \\
&+ 48{g_1}^4 \\
&- 30{g_1}^2\lambda_{12}^{(1)} \\
&+ 4{\lambda_{12}^{(1)}}^2 \\
&+ \lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
&+ \lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
&+ \lambda_{12}^{(1)}(4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})
\end{aligned}
$$

$$
\begin{aligned}
&+ 16{\lambda_{12}^{(A)}}^2 \\
&+ 4{\lambda_{12}^{(\times)}}^2 \\
&+ 4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ 4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_5}{d\ln\mu} = \lambda_5 &\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3}\right) \\
&- \frac{33{g_2}^2\lambda_5}{2} \\
&- \frac{33{g_1}^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_1}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} = \lambda_{HS_1}^{(A)}&(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda_{S_1}^{(A)}) \\
&- \frac{4}{3}\lambda_5\,(\lambda_5)^* \\
&+ {g_1}^2\left(-8{g_2}^2 - \frac{51\lambda_{HS_1}^{(A)}}{2}\right) \\
&- \frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2} \\
&+ 12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = \lambda_{HS_2}^{(A)}&(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda_{S_2}^{(A)}) \\
&- \frac{8}{9}\lambda_5\,(\lambda_5)^* \\
&+ {g_1}^2\left(-4{g_2}^2 - \frac{15\lambda_{HS_2}^{(A)}}{2}\right) \\
&- \frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}
\end{aligned}
$$

$$
\begin{aligned}
&+ 12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{S_1}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(A)}}{d\ln\mu} = &-2{g_2}^4 \\
&+ {g_1}^2\left(32{g_2}^2 - 48\lambda_{S_1}^{(A)}\right) \\
&- 24{g_2}^2\lambda_{S_1}^{(A)} \\
&+ 12{\lambda_{12}^{(A)}}^2 \\
&- \frac{4{\lambda_{12}^{(\times)}}^2}{3} \\
&+ 3{\lambda_{HS_1}^{(A)}}^2 \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 20{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(A)}}{d\ln\mu} = &-2{g_2}^4 \\
&+ {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_2}^{(A)}\right) \\
&- 24{g_2}^2\lambda_{S_2}^{(A)} \\
&+ 12{\lambda_{12}^{(A)}}^2 \\
&- \frac{4{\lambda_{12}^{(\times)}}^2}{3} \\
&+ 3{\lambda_{HS_2}^{(A)}}^2 \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(A)}}{d\ln\mu} = &-\frac{2}{9}\lambda_5\left(\lambda_5\right)^* \\
&+ {g_1}^2\left(16{g_2}^2 - 30\lambda_{12}^{(A)}\right) \\
&- 24{g_2}^2\lambda_{12}^{(A)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ \frac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)} \\
&+ 3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}
\end{aligned}
$$

$\lambda_{12}^{(\times)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(\times)}}{d\ln\mu} = &-\lambda_5 \left(\lambda_5\right)^* \\
&+ \frac{51{g_2}^4}{10} \\
&- 24{g_2}^2\lambda_{12}^{(\times)} \\
&- 30{g_1}^2\lambda_{12}^{(\times)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ 10{\lambda_{12}^{(\times)}}^2 \\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)} \\
&- 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)} \\
&- 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}
\end{aligned}
$$